

Day 3: Data Sources & Storage Modes

📌 **Definition to Remember**

Power BI connects to over 150+ data sources using three primary storage modes: **Import Mode** (in-memory VertiPaq compression), **DirectQuery** (live source pass-through), and **Dual/Composite Mode** (hybrid).

1. Common Data Sources

- **Flat Files:** Excel (`.xlsx`), CSV (`.csv`), JSON, XML.
- **Folder Connector:** Ingests and combines dozens of identical monthly/daily files into a single unified table automatically.
- **Relational Databases:** SQL Server, Oracle, PostgreSQL, MySQL, IBM Db2.
- **Web & REST APIs:** Pulling live web tables and JSON endpoints with API authentication keys.

2. Storage Modes Compared

Feature	Import Mode	DirectQuery Mode	Composite / Dual Mode
Data Location	Loaded into memory (RAM) via VertiPaq engine.	Remains at the underlying source (SQL/Snowflake).	Tables set to Import or DirectQuery as needed.
Performance	Blazing fast columnar compression and in-memory cache.	Dependent on backend database speed and network latency.	Optimized: High-speed aggregations with live drilldown.
DAX Capability	Full, unrestricted DAX function support.	Limited DAX (must translate into native SQL queries).	Full DAX on import tables; translated on DirectQuery.
Data Freshness	Requires scheduled refresh.	Real-time live data on every visual render.	Near real-time with scheduled aggregations.
Dataset Limit	Up to 1 GB (Pro) or 400 GB (Premium).	No Power BI size limit (handles multi-terabyte data).	Combines massive tables with cached dimensions.

★ **Must-Master Skills & Interview Points**

1. Why Import Mode is preferred in 90% of BI projects (VertiPaq speed and full DAX support).
2. When DirectQuery is mandatory: strict regulatory security, real-time data needs, or massive petabyte datasets.
3. How the Folder connector eliminates manual copy-pasting for recurring data drops.

⚡ Quick Recall

Import (Fastest, in-memory, scheduled refresh) vs DirectQuery (Live SQL pass-through, real-time, query limits)